# Assignment 2 Report

Beta, Cost of Equity, Cost of Debt and WACC: The Coca-Cola Company (KO) vs. Amazon.com (AMZN)

## Data and Method

I used 61 month-end adjusted closing prices (October 2021 to October 2026) for KO, AMZN and the S&P 500 (^GSPC), giving 60 monthly returns over five years. Returns are R(t) = P(t)/P(t−1) − 1. Prices come from Digrin's monthly price history (Yahoo Finance-based data); the S&P 500 is the index level and so excludes dividends. All calculations are live formulas in the accompanying Excel workbook.

## Question 1: Beta Estimates

Each stock's return was regressed on the S&P 500 return: R(i) = α + β·R(m) + ε.

| | KO | AMZN |
|---|---|---|
| **Beta (β)** | 0.281 | 1.495 |
| **Std. error of β** | 0.135 | 0.213 |
| **95% CI for β** | 0.011 to 0.552 | 1.069 to 1.921 |
| **Alpha α (monthly)** | 0.95% | -0.34% |
| **$R^2$** | 7.0% | 45.9% |
| **Annualized volatility** | 16.4% | 34.0% |

Interpretation: β(KO) = 0.281 means KO moves only about 0.28% for each 1% move in the market, a defensive stock. β(AMZN) = 1.495 means AMZN amplifies market moves by roughly 1.5 times. Market movements explain just 7.0% of KO's return variation but 45.9% of AMZN's. Neither alpha is statistically different from zero.

## Question 2: Cost of Equity (CAPM)

r(e) = r(f) + β·[E(r(m)) − r(f)]. I used r(f) = 4.9% (the 10-year Treasury yield was about 4.9% in early November 2026, not the 4.5% example in the prompt) and a market risk premium of 5.5%, the midpoint of the suggested 5–6% range. This gives r(e) = 6.45% for KO and 13.12% for AMZN, a gap of about 6.7 percentage points driven entirely by beta. The workbook includes a sensitivity table for premiums of 5%, 5.5% and 6%.

## Question 3: Cost of Debt

Using FY2025 Form 10-K data, r(d) = interest expense ÷ average total debt. KO: $1,654M ÷ $45,007M = 3.67%. AMZN: $2,274M ÷ $59,136M = 3.85%. Tax rates are 20.7% for KO (underlying effective rate) and 19.6% for AMZN (provision ÷ pre-tax income), so after-tax costs of debt are 2.91% and 3.09%.

## Question 4: WACC

Equity is valued at market (latest price × shares outstanding from the most recent 10-Qs); debt is FY2025 book value as a proxy for market value. WACC = (E/V)·r(e) + (D/V)·r(d)(1 − Tc).

| Firm | E | D | E/V : D/V | After-tax r(d) / r(e) | WACC |
|---|---|---|---|---|---|
| **KO** | $413B | $45B | 90.1% : 9.9% | 2.91% / 6.45% | 6.10% |
| **AMZN** | $2,672B | $66B | 97.6% : 2.4% | 3.09% / 13.12% | 12.88% |

## Question 5: Discussion

AMZN carries far higher systematic risk and cost of equity. Its estimated beta is 1.495 versus 0.281 for KO, which gives CAPM costs of equity of 13.1% and 6.4% respectively. Amazon's earnings depend on consumer spending, advertising and corporate cloud and AI investment, all of which weaken in downturns, and its value rests on distant cash flows that are very sensitive to discount rates. Coca-Cola sells a staple whose demand holds up in recessions. KO's beta estimate is imprecise ($R^2$ of only 7.0%; 95% interval 0.011 to 0.552), but even its upper bound implies a cost of equity well below AMZN's.

Capital structure matters because debt is cheaper than equity: lenders have a senior claim and interest is tax-deductible, so each dollar of debt replaces costlier equity and lowers WACC, up to the point where distress costs begin to bite. KO uses more leverage (D/V 9.9%) than AMZN (D/V 2.4%), yet KO's WACC of 6.1% is low mainly because its equity is cheap, not because of its debt. AMZN's WACC of 12.9% is almost identical to its cost of equity because debt is such a small share of capital. Note also that a firm's equity beta already reflects its leverage, so KO's underlying business risk is lower still.

These numbers say a lot about the firms. AMZN's 12.9% hurdle rate reflects a growth company whose value comes from large future investment opportunities; each new project must earn well above that rate. KO is a mature, stable-cash-flow business whose projects only need to clear about 6.1%. Business risk is therefore much higher at AMZN, while its small debt load and enormous equity value give it great financial flexibility. KO's A+/A1 credit ratings give it cheap, reliable access to debt. Investors in KO expect a modest return, largely from dividends and steady growth, whereas AMZN investors demand roughly double that through capital gains.

This connects directly to Assignment 1. A stock's total variance splits into a systematic part, which cannot be diversified away, and an idiosyncratic part, which can. KO's annualized volatility (16.4%) is close to the market's (15.4%), but its correlation with the market is only 0.264, so about 93% of its variance is diversifiable and its beta is low. AMZN's correlation is 0.678 and its volatility is 34.0%, so 46% of its variance is systematic. Because beta equals correlation times $\sigma(\text{stock})/\sigma(\text{market})$, CAPM prices only the part of risk that diversification cannot remove.

## Assumptions and Limitations

The risk-free rate and market risk premium are assumptions. The S&P 500 index excludes dividends. The five-year window is short, so KO's beta in particular has a wide confidence interval. AMZN debt includes long-term debt only (not the current portion or leases) while its interest expense includes finance-lease interest, so its cost of debt is approximate. Neither firm's current yield to maturity was used. The source data shows the same KO price for November and December 2022, which I kept as published.